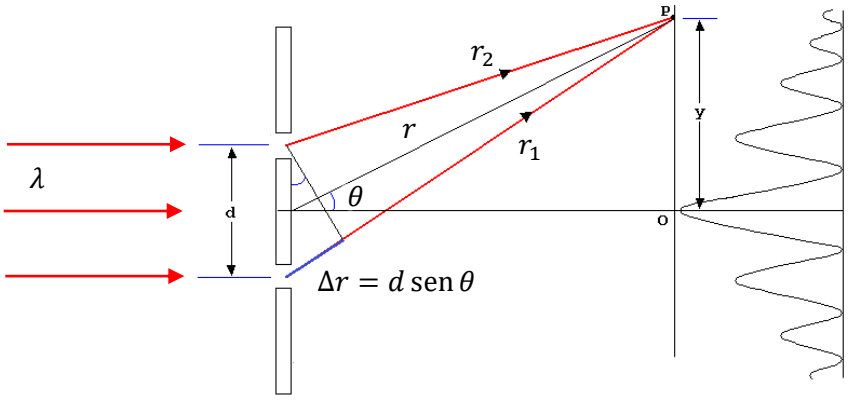
La doble rendija original



Las expresiones para los rayos que salen de las rendijas

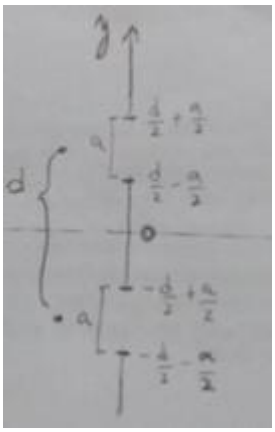
$$E_{\mathbb{C}1}(y,t) = A\, e^{i(kr_1-\omega t)}$$
$$E_{\mathbb{C}2}(y,t) = A\, e^{i(kr_2-\omega t)}$$

r_1 y r_2 son funciones de y

$$r_1 = r - |y|.\operatorname{sen}\theta = r + y.\operatorname{sen}\theta \quad (y<0)$$
$$r_2 = r + |y|.\operatorname{sen}\theta = r + y.\operatorname{sen}\theta \quad (y>0)$$

$$E_{\mathbb{C}1}(y,t) = E_{\mathbb{C}2}(y,t) = A\, e^{i(kr+ky\operatorname{sen}\theta-\omega t)} = E_{\mathbb{C}}(y,t)$$

$$\phi = kr - \omega t$$
$$E_{\mathbb{C}}(y,t) = A\, e^{i\phi}\, e^{iky\operatorname{sen}\theta}$$



Ahora sumamos todos los rayos que salen de cada punto del interior de las rendijas

$$E_{\mathbb{C}}(y,t) = \int_{-\frac{d}{2}-\frac{a}{2}}^{-\frac{d}{2}+\frac{a}{2}} E_{\mathbb{C}}\, dy + \int_{\frac{d}{2}-\frac{a}{2}}^{\frac{d}{2}+\frac{a}{2}} E_{\mathbb{C}}\, dy$$

$$E_{\mathbb{C}}(y,t) = Ae^{i\phi} \int_{-\frac{d}{2}-\frac{a}{2}}^{-\frac{d}{2}+\frac{a}{2}} e^{i(k\operatorname{sen}\theta)y}\, dy + Ae^{i\phi} \int_{-\frac{d}{2}-\frac{a}{2}}^{-\frac{d}{2}+\frac{a}{2}} e^{i(k\operatorname{sen}\theta)y}\, dy$$

Definimos $u \equiv k\operatorname{sen}\theta$

$$E_{\mathbb{C}}(y,t) = Ae^{i\phi} \int_{-\frac{d}{2}-\frac{a}{2}}^{-\frac{d}{2}+\frac{a}{2}} e^{iuy}\, dy + Ae^{i\phi} \int_{-\frac{d}{2}-\frac{a}{2}}^{-\frac{d}{2}+\frac{a}{2}} e^{iuy}\, dy$$

$$E_{\mathbb{C}}(y,t) = Ae^{i\phi}\, \frac{e^{iuy}}{iu}\Bigg|_{-\frac{d}{2}-\frac{a}{2}}^{-\frac{d}{2}+\frac{a}{2}} + Ae^{i\phi}\, \frac{e^{iuy}}{iu}\Bigg|_{\frac{d}{2}-\frac{a}{2}}^{\frac{d}{2}+\frac{a}{2}}$$

$$E_{\mathbb{C}}(y,t) = \frac{Ae^{i\phi}}{iu}\left[e^{iu\left(-\frac{d}{2}+\frac{a}{2}\right)} - e^{iu\left(-\frac{d}{2}-\frac{a}{2}\right)}\right] + \frac{Ae^{i\phi}}{iu}\left[e^{iu\left(\frac{d}{2}+\frac{a}{2}\right)} - e^{iu\left(\frac{d}{2}-\frac{a}{2}\right)}\right]$$

$$E_{\mathbb{C}}(y,t) = \frac{Ae^{i\phi}}{iu}e^{-iu\frac{d}{2}}\left[e^{iu\frac{a}{2}} - e^{-iu\frac{a}{2}}\right] + \frac{Ae^{i\phi}}{iu}e^{iu\frac{d}{2}}\left[e^{iu\frac{a}{2}} - e^{-iu\frac{a}{2}}\right]$$

$$E_{\mathbb{C}}(y,t) = \frac{Ae^{i\phi}}{iu}\left[e^{-iu\frac{d}{2}} + e^{iu\frac{d}{2}}\right]\left[e^{iu\frac{a}{2}} - e^{-iu\frac{a}{2}}\right]$$

$$E_{\mathbb{C}}(y,t) = \frac{Ae^{i\phi}}{iu}\ 2\cos\left(u\frac{d}{2}\right)\ 2i\,\mathrm{sen}\left(u\frac{a}{2}\right)$$

$$E_{\mathbb{C}}(y,t) = Ae^{i\phi}\ 2\cos\left(u\frac{d}{2}\right)\ \frac{\frac{a}{2}\,2i\,\mathrm{sen}\left(u\frac{a}{2}\right)}{iu\,\frac{a}{2}}$$

Definimos $\beta \equiv u\frac{a}{2} = (k\,\mathrm{sen}\,\theta)\frac{a}{2}$; $\gamma \equiv u\frac{d}{2} = (k\,\mathrm{sen}\,\theta)\frac{d}{2}$

$$E_{\mathbb{C}}(y,t) = Aa\ e^{i\phi}\ 2\cos(\gamma)\ \frac{\mathrm{sen}(\beta)}{\beta}$$

La parte real: $E = Aa\cos(\phi)\ 2\cos\gamma\left(\frac{\mathrm{sen}\,\beta}{\beta}\right)$

ϕ es una función de t que varía muy rápido

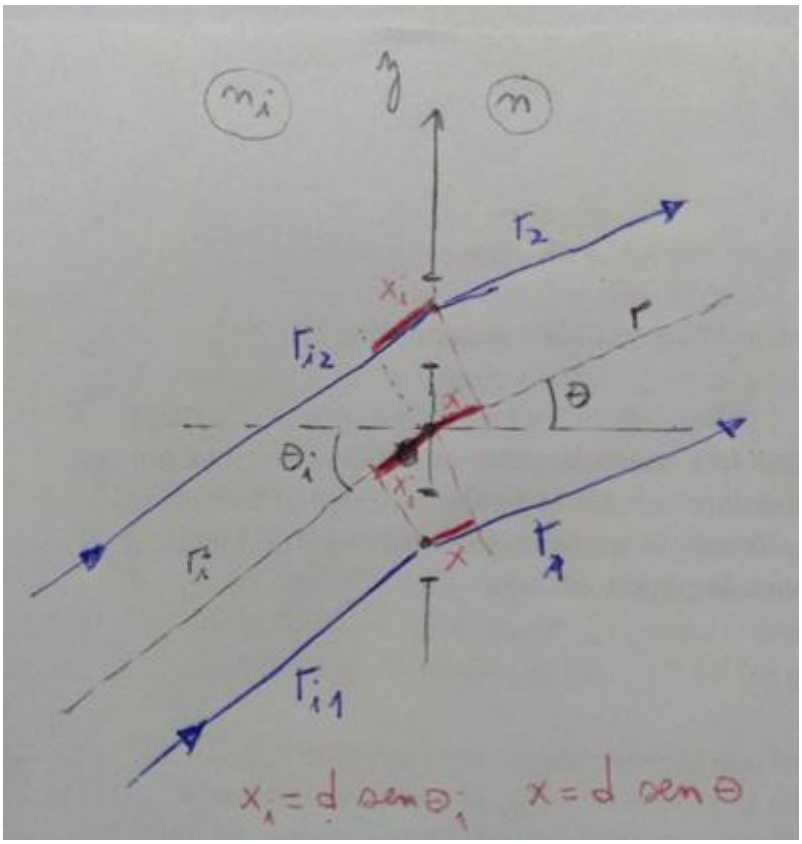
$$I = nc\varepsilon_0\langle|E|^2\rangle = \frac{nc\varepsilon_0}{2}(Aa)^2\ 4\cos^2\gamma\left(\frac{\mathrm{sen}\,\beta}{\beta}\right)^2$$

$$\boxed{I = 4I_0\cos^2\gamma\left(\frac{\mathrm{sen}\,\beta}{\beta}\right)^2}$$

$$\gamma = \frac{d}{2}k\,\mathrm{sen}\,\theta = \frac{\pi d}{\lambda_0}n\,\mathrm{sen}\,\theta = \frac{\pi}{\lambda}d\,\mathrm{sen}\,\theta$$

$$\beta = \frac{a}{2}k\,\mathrm{sen}\,\theta = \frac{\pi a}{\lambda_0}n\,\mathrm{sen}\,\theta = \frac{\pi}{\lambda}a\,\mathrm{sen}\,\theta$$

Se hace incidir ahora la luz en forma oblicua con un ángulo θ_i



Para los rayos antes de llegar a las rendijas

$$E_{\mathbb{C}i1}(y,t) = A\, e^{i(k_i r_{i1} - \omega t_i)}$$
$$E_{\mathbb{C}i2}(y,t) = A\, e^{i(k_i r_{i2} - \omega t_i)}$$

r_{i1} y r_{i2} son funciones de y

$$r_{i1} = r_i - |y|.\operatorname{sen}\theta_i = r_i + y.\operatorname{sen}\theta_i \quad (y<0)$$
$$r_{i2} = r_i + |y|.\operatorname{sen}\theta_i = r_i + y.\operatorname{sen}\theta_i \quad (y>0)$$
$$E_{\mathbb{C}i1}(y,t) = E_{\mathbb{C}i2}(y,t) = A\, e^{i(k_i r_i + k_i y \operatorname{sen}\theta_i - \omega t_i)} = E_{\mathbb{C}i}(y,t)$$

$$\phi_i = k_i r_i - \omega t_i$$
$$E_{\mathbb{C}i}(y,t) = A\, e^{i\phi_i}\, e^{ik_i y \operatorname{sen}\theta_i}$$

Para los rayos que van de las rendijas a la pantalla

$$E_{\mathbb{C}1}(r,t) = A\, e^{i\phi_i}\, e^{ik_i y \operatorname{sen}\theta_i}\, e^{i(kr_1 - \omega t)}$$
$$E_{\mathbb{C}2}(r,t) = A\, e^{i\phi_i}\, e^{ik_i y \operatorname{sen}\theta_i}\, e^{i(kr_2 - \omega t)}$$

r_1 y r_2 son funciones de y

$$r_1 = r + |y|.\operatorname{sen}\theta = r - y.\operatorname{sen}\theta \quad (y<0)$$
$$r_2 = r - |y|.\operatorname{sen}\theta = r - y.\operatorname{sen}\theta \quad (y>0)$$

$$E_{\mathbb{C}1}(y,t) = E_{\mathbb{C}2}(y,t) = A\, e^{i\phi_i}\, e^{ik_i y \operatorname{sen}\theta_i}\, e^{i(kr - ky\operatorname{sen}\theta - \omega t)} = E_{\mathbb{C}f}(y,t)$$
$$\phi_f = kr - \omega t$$
$$E_{\mathbb{C}f}(y,t) = A\, e^{i\phi_i}\, e^{ik_i y \operatorname{sen}\theta_i}\, e^{i\phi_f}\, e^{-iky\operatorname{sen}\theta}$$
$$E_{\mathbb{C}f}(y,t) = A\, e^{i(\phi_i + \phi_f)}\, e^{i(k_i \operatorname{sen}\theta_i - k\operatorname{sen}\theta)y} = A\, e^{i\phi}\, e^{i(k_i \operatorname{sen}\theta_i - k\operatorname{sen}\theta)y}$$

Ahora sumamos todos los rayos que salen de cada punto del interior de las rendijas

$$E_{\mathbb{C}}(y,t) = \int_{-\frac{d}{2}-\frac{a}{2}}^{-\frac{d}{2}+\frac{a}{2}} E_{\mathbb{C}}\, dy + \int_{\frac{d}{2}-\frac{a}{2}}^{\frac{d}{2}+\frac{a}{2}} E_{\mathbb{C}}\, dy$$

$$E_{\mathbb{C}}(y,t) = Ae^{i\phi} \int_{-\frac{d}{2}-\frac{a}{2}}^{-\frac{d}{2}+\frac{a}{2}} e^{i(k_i \operatorname{sen}\theta_i - k\operatorname{sen}\theta)y}\, dy + Ae^{i\phi} \int_{-\frac{d}{2}-\frac{a}{2}}^{-\frac{d}{2}+\frac{a}{2}} e^{i(k_i \operatorname{sen}\theta_i - k\operatorname{sen}\theta)y}\, dy$$

Definimos $u \equiv k_i \operatorname{sen}\theta_i - k\operatorname{sen}\theta$

$$E_{\mathbb{C}}(y,t) = Ae^{i\phi} \int_{-\frac{d}{2}-\frac{a}{2}}^{-\frac{d}{2}+\frac{a}{2}} e^{iuy}\, dy + Ae^{i\phi} \int_{-\frac{d}{2}-\frac{a}{2}}^{-\frac{d}{2}+\frac{a}{2}} e^{iuy}\, dy$$

$$E_{\mathbb{C}}(y,t) = Ae^{i\phi}\, \frac{e^{iuy}}{iu}\Bigg|_{-\frac{d}{2}-\frac{a}{2}}^{-\frac{d}{2}+\frac{a}{2}} + Ae^{i\phi}\, \frac{e^{iuy}}{iu}\Bigg|_{\frac{d}{2}-\frac{a}{2}}^{\frac{d}{2}+\frac{a}{2}}$$

$$E_{\mathbb{C}}(y,t) = \frac{Ae^{i\phi}}{iu}\left[e^{iu\left(-\frac{d}{2}+\frac{a}{2}\right)} - e^{iu\left(-\frac{d}{2}-\frac{a}{2}\right)}\right] + \frac{Ae^{i\phi}}{iu}\left[e^{iu\left(\frac{d}{2}+\frac{a}{2}\right)} - e^{iu\left(\frac{d}{2}-\frac{a}{2}\right)}\right]$$

$$E_{\mathbb{C}}(y,t) = \frac{Ae^{i\phi}}{iu} e^{-iu\frac{d}{2}}\left[e^{iu\frac{a}{2}} - e^{-iu\frac{a}{2}}\right] + \frac{Ae^{i\phi}}{iu} e^{iu\frac{d}{2}}\left[e^{iu\frac{a}{2}} - e^{-iu\frac{a}{2}}\right]$$

$$E_{\mathbb{C}}(y,t) = \frac{Ae^{i\phi}}{iu}\left[e^{-iu\frac{d}{2}} + e^{iu\frac{d}{2}}\right]\left[e^{iu\frac{a}{2}} - e^{-iu\frac{a}{2}}\right]$$

$$E_{\mathbb{C}}(y,t) = \frac{Ae^{i\phi}}{iu}\ 2\cos\left(u\tfrac{d}{2}\right)\ 2i\,\mathrm{sen}\left(u\tfrac{a}{2}\right)$$

$$E_{\mathbb{C}}(y,t) = Ae^{i\phi}\ 2\cos\left(u\tfrac{d}{2}\right)\ \frac{\frac{a}{2}\,2i\,\mathrm{sen}\left(u\frac{a}{2}\right)}{iu\,\frac{a}{2}}$$

Definimos $(-\beta) \equiv u\frac{a}{2} = (k_i \,\mathrm{sen}\,\theta_i - k\,\mathrm{sen}\,\theta)\frac{a}{2}$; $(-\gamma) \equiv u\frac{d}{2} = (k_i \,\mathrm{sen}\,\theta_i - k\,\mathrm{sen}\,\theta)\frac{d}{2}$

$$E_{\mathbb{C}}(y,t) = Aa\ e^{i\phi}\ 2\cos(-\gamma)\ \frac{\mathrm{sen}(-\beta)}{-\beta}$$

La parte real: $E = Aa\cos(\phi)\ 2\cos\gamma\left(\frac{-\,\mathrm{sen}\,\beta}{-\beta}\right) = Aa\cos(\phi)\ 2\cos\gamma\left(\frac{\mathrm{sen}\,\beta}{\beta}\right)$

ϕ es una función de t que varía muy rápido

$$I = nc\varepsilon_0\langle|E|^2\rangle = \frac{nc\varepsilon_0}{2}(Aa)^2\ 4\cos^2\gamma\left(\frac{\mathrm{sen}\,\beta}{\beta}\right)^2$$

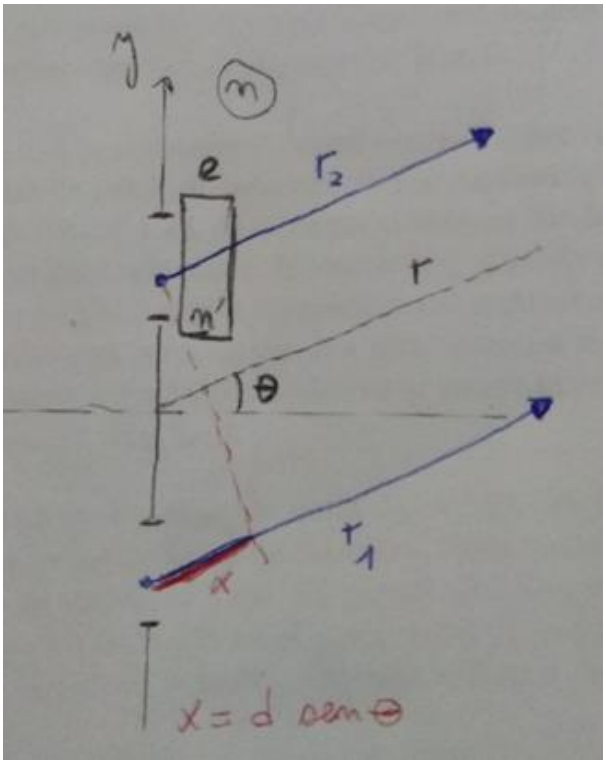
$$\boxed{I = 4I_0\cos^2\gamma\left(\frac{\mathrm{sen}\,\beta}{\beta}\right)^2}$$

$$\gamma = \frac{d}{2}(k\,\mathrm{sen}\,\theta - k_i\,\mathrm{sen}\,\theta_i) = \frac{\pi d}{\lambda_0}[n\,\mathrm{sen}\,\theta - n_i\,\mathrm{sen}\,\theta_i]$$

$$\beta = \frac{a}{2}(k\,\mathrm{sen}\,\theta - k_i\,\mathrm{sen}\,\theta_i) = \frac{\pi a}{\lambda_0}[n\,\mathrm{sen}\,\theta - n_i\,\mathrm{sen}\,\theta_i]$$

El término adicional en γ muestra que los máximos de interferencia se desplazan como consecuencia de la incidencia oblicua. Como β tiene la misma forma que γ, la campana de difracción se desplaza igual que los máximos de interferencia.

Con una lámina delgada colocada luego de la rendija superior



$$E_{\mathbb{C}1}(y,t) = A\, e^{i(kr_1-\omega t)}$$
$$E_{\mathbb{C}2}(y,t) = A\, e^{i(k'e+k(r_2-e)-\omega t)}$$

r_1 y r_2 son funciones de y

$$r_1 = r + |y|.\operatorname{sen}\theta = r - y.\operatorname{sen}\theta \quad (y<0)$$
$$r_2 = r - |y|.\operatorname{sen}\theta = r - y.\operatorname{sen}\theta \quad (y>0)$$
$$E_{\mathbb{C}1}(y,t) = A\, e^{i(kr-ky\operatorname{sen}\theta-\omega t)}$$
$$E_{\mathbb{C}2}(y,t) = A\, e^{i(k'e+k(r-y\operatorname{sen}\theta-e)-\omega t)}$$

$$\phi_1 = kr-\omega t \;\; ; \;\; \phi_2 = kr + k'e - ke - \omega t = \phi_1 + (k'-k)e$$
$$E_{\mathbb{C}1}(y,t) = A\, e^{i\phi_1}\, e^{-iky\operatorname{sen}\theta}$$
$$E_{\mathbb{C}2}(y,t) = A\, e^{i\phi_2}\, e^{-iky\operatorname{sen}\theta}$$

$$E_{\mathbb{C}}(y,t) = \int_{-\frac{d}{2}-\frac{a}{2}}^{-\frac{d}{2}+\frac{a}{2}} E_{\mathbb{C}1}(y,t)\,dy + \int_{\frac{d}{2}-\frac{a}{2}}^{\frac{d}{2}+\frac{a}{2}} E_{\mathbb{C}2}(y,t)\,dy$$

$$E_{\mathbb{C}}(y,t) = A\, e^{i\phi_1}\int_{-\frac{d}{2}-\frac{a}{2}}^{-\frac{d}{2}+\frac{a}{2}} e^{-iky\operatorname{sen}\theta}dy + A\, e^{i\phi_2}\int_{\frac{d}{2}-\frac{a}{2}}^{\frac{d}{2}+\frac{a}{2}} e^{-iky\operatorname{sen}\theta}dy$$

Definimos $u = k\operatorname{sen}\theta$ y resolvemos la primera integral

$$A\, e^{i\phi_1}\int_{-\frac{d}{2}-\frac{a}{2}}^{-\frac{d}{2}+\frac{a}{2}} e^{-iuy}dy = A\, e^{i\phi_1}\frac{e^{-iuy}}{-iu}\Bigg|_{-\frac{d}{2}-\frac{a}{2}}^{-\frac{d}{2}+\frac{a}{2}} =$$

$$= \frac{A\, e^{i\phi_1}}{-iu}\left[e^{iu\left(\frac{d}{2}-\frac{a}{2}\right)} - e^{iu\left(\frac{d}{2}+\frac{a}{2}\right)}\right] = \frac{A\, e^{i\phi_1}}{-iu}\, e^{iu\frac{d}{2}}\left(e^{-iu\frac{a}{2}} - e^{iu\frac{a}{2}}\right) = \frac{A\, e^{i\phi_1}}{-iu}\, e^{iu\frac{d}{2}}\,(-2i)\operatorname{sen}\left(u\frac{a}{2}\right)$$
$$= A\, e^{i\phi_1}\, e^{iu\frac{d}{2}}\,\frac{\frac{a}{2}\,2\operatorname{sen}\left(u\frac{a}{2}\right)}{u\frac{a}{2}}$$

Definimos $\beta \equiv u\frac{a}{2} = k\frac{a}{2}\operatorname{sen}\theta$

$$A\, e^{i\phi_1}\int_{-\frac{d}{2}-\frac{a}{2}}^{-\frac{d}{2}+\frac{a}{2}} e^{-iuy}dy = Aa\, e^{i\left(\phi_1+u\frac{d}{2}\right)}\left(\frac{\operatorname{sen}\beta}{\beta}\right)$$

Ahora la segunda integral

$$A\, e^{i\phi_2}\int_{\frac{d}{2}-\frac{a}{2}}^{\frac{d}{2}+\frac{a}{2}} e^{-iuy}dy = A\, e^{i\phi_2}\frac{e^{-iuy}}{-iu}\Bigg|_{\frac{d}{2}-\frac{a}{2}}^{\frac{d}{2}+\frac{a}{2}} =$$

$$= \frac{A\, e^{i\phi_2}}{-iu}\left[e^{-iu\left(\frac{d}{2}+\frac{a}{2}\right)} - e^{-iu\left(\frac{d}{2}-\frac{a}{2}\right)}\right] = \frac{A\, e^{i\phi_2}}{-iu}\, e^{-iu\frac{d}{2}}\left(e^{-iu\frac{a}{2}} - e^{iu\frac{a}{2}}\right) = \frac{A\, e^{i\phi_2}}{-iu}\, e^{-iu\frac{d}{2}}\,(-2i)\operatorname{sen}\left(u\frac{a}{2}\right)$$

$$A\, e^{i\phi_2}\int_{\frac{d}{2}-\frac{a}{2}}^{\frac{d}{2}+\frac{a}{2}} e^{-iuy}dy = Aa\, e^{i\left(\phi_2-u\frac{d}{2}\right)}\left(\frac{\operatorname{sen}\beta}{\beta}\right)$$

$$E_{\mathbb{C}} = Aa\left(\frac{\operatorname{sen}\beta}{\beta}\right)\left[e^{i\phi_1}\, e^{iu\frac{d}{2}} + e^{i\phi_2}\, e^{-iu\frac{d}{2}}\right]$$

$$E_{\mathbb{C}} = Aa\left(\frac{\operatorname{sen}\beta}{\beta}\right)e^{i\phi_1}\left[\, e^{iu\frac{d}{2}} + e^{i[\phi_1+(k'-k)e]}\, e^{-iu\frac{d}{2}}\right]$$

$$E_{\mathbb{C}} = Aa\left(\frac{\operatorname{sen}\beta}{\beta}\right)e^{i\phi_1}\left[\, e^{iu\frac{d}{2}} + e^{i(k'-k)e}\, e^{-iu\frac{d}{2}}\right]$$

$$E_{\mathbb{C}} = Aa\left(\frac{\operatorname{sen}\beta}{\beta}\right)e^{i\phi_1}e^{-i(k'-k)\frac{e}{2}}\left[\, e^{i(k'-k)\frac{e}{2}}\, e^{iu\frac{d}{2}} + e^{-i(k'-k)\frac{e}{2}}\, e^{-iu\frac{d}{2}}\right]$$

$$E_{\mathbb{C}} = Aa\left(\frac{\operatorname{sen}\beta}{\beta}\right)e^{i\phi}\left[\, e^{i(k'-k)\frac{e}{2}}\, e^{iu\frac{d}{2}} + e^{-i(k'-k)\frac{e}{2}}\, e^{-iu\frac{d}{2}}\right]$$

$$E_{\mathbb{C}} = Aa\left(\frac{\operatorname{sen}\beta}{\beta}\right)e^{i\phi}\, 2\cos\left[u\frac{d}{2} + (k'-k)\frac{e}{2}\right]$$

$$E_{\mathbb{C}} = Aa\, e^{i\phi}\, 2\cos\left[k\frac{d}{2}\operatorname{sen}\theta + (k'-k)\frac{e}{2}\right]\left(\frac{\operatorname{sen}\beta}{\beta}\right)$$

La parte real: $E = Aa\cos\phi\ \ 2\cos\left[k\frac{d}{2}\operatorname{sen}\theta + (k'-k)\frac{e}{2}\right]\left(\frac{\operatorname{sen}\beta}{\beta}\right)$

ϕ es una función de t que varía muy rápido

$$I = nc\varepsilon_0\langle|E|^2\rangle = \frac{nc\varepsilon_0}{2}(Aa)^2\, 4\cos^2\left[k\frac{d}{2}\operatorname{sen}\theta + (k'-k)\frac{e}{2}\right]\left(\frac{\operatorname{sen}\beta}{\beta}\right)^2$$

$$\boxed{I = 4I_0\cos^2\gamma\left(\frac{\operatorname{sen}\beta}{\beta}\right)^2}$$

$$\gamma \equiv k\frac{d}{2}\operatorname{sen}\theta + (k'-k)\frac{e}{2} = \frac{\pi}{\lambda_0}[dn\operatorname{sen}\theta + (n'-n)e]$$

$$\beta = k\frac{a}{2}\operatorname{sen}\theta = \frac{\pi}{\lambda_0}an\operatorname{sen}\theta$$

La expresión de γ tiene términos adicionales, mostrando que los máximos de interferencia se desplazan al colocar la lámina. Como β tiene la misma expresión que al principio, la lámina no provoca desplazamiento de la campana de difracción.